All India Aakash Test Series for NEET - 2025

TEST - 1 (Code-E)

Test Date : 10/11/2024

ANSWERS

1.	(2)	41.	(3)	81.	(2)	121.	(3)	161.	(2)
2.	(3)	42.	(4)	82.	(3)	122.	(4)	162.	(4)
3.	(1)	43.	(2)	83.	(2)	123.	(4)	163.	(2)
4.	(4)	44.	(1)	84.	(3)	124.	(3)	164.	(3)
5.	(2)	45.	(4)	85.	(3)	125.	(4)	165.	(2)
6.	(3)	46.	(2)	86.	(2)	126.	(4)	166.	(4)
7.	(4)	47.	(2)	87.	(4)	127.	(2)	167.	(3)
8.	(1)	48.	(3)	88.	(3)	128.	(2)	168.	(1)
9.	(2)	49.	(4)	89.	(4)	129.	(4)	169.	(4)
10.	(4)	50.	(1)	90.	(4)	130.	(3)	170.	(2)
11.	(2)	51.	(2)	91.	(1)	131.	(4)	171.	(3)
12.	(2)	52.	(1)	92.	(4)	132.	(2)	172.	(3)
13.	(1)	53.	(4)	93.	(2)	133.	(2)	173.	(4)
14.	(4)	54.	(2)	94.	(3)	134.	(4)	174.	(3)
15.	(2)	55.	(3)	95.	(2)	135.	(2)	175.	(1)
16.	(3)	56.	(2)	96.	(4)	136.	(3)	176.	(3)
17.	(2)	57.	(4)	97.	(3)	137.	(4)	177.	(4)
18.	(1)	58.	(1)	98.	(4)	138.	(3)	178.	(3)
19.	(1)	59.	(4)	99.	(3)	139.	(3)	179.	(1)
20.	(3)	60.	(2)	100.	(1)	140.	(1)	180.	(4)
21.	(1)	61.	(2)	101.	(2)	141.	(4)	181.	(3)
22.	(4)	62.	(3)	102.	(3)	142.	(1)	182.	(2)
23.	(2)	63.	(1)	103.	(3)	143.	(4)	183.	(1)
24.	(4)	64.	(4)	104.	(3)	144.	(3)	184.	(2)
25.	(4)	65.	(3)	105.	(4)	145.	(4)	185.	(2)
26.	(4)	66.	(1)	106.	(4)	146.	(1)	186.	(3)
27.	(3)	67.	(2)	107.	(3)	147.	(3)	187.	(4)
28.	(2)	68.	(1)	108.	(4)	148.	(1)	188.	(1)
29.	(3)	69.	(3)	109.	(3)	149.	(2)	189.	(4)
30.	(3)	70.	(2)	110.	(3)	150.	(3)	190.	(3)
31.	(4)	71.	(3)	111.	(4)	151.	(3)	191.	(4)
32.	(2)	72.	(2)	112.	(2)	152.	(2)	192.	(1)
33.	(2)	73.	(3)	113.	(4)	153.	(4)	193.	(2)
34.	(2)	74.	(1)	114.	(2)	154.	(1)	194.	(2)
35.	(1)	75.	(4)	115.	(1)	155.	(2)	195.	(3)
36.	(3)	76.	(1)	116.	(2)	156.	(1)	196.	(4)
37.	(1)	77.	(3)	117.	(3)	157.	(2)	197.	(2)
38.	(3)	78.	(4)	118.	(2)	158.	(3)	198.	(3)
39.	(1)	79.	(3)	119.	(4)	159.	(3)	199.	(1)
40.	(3)	80.	(1)	120.	(4)	160.	(4)	200.	(2)

HINTS & SOLUTIONS

[PHYSICS]

SECTION-A

1. Answer (2)

Hint & Sol.: $y = a \times b$

$$\frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b}$$

$$\frac{\Delta y}{y} \times 100 = \frac{\Delta a}{a} \times 100 + \frac{\Delta b}{b} \times 100$$

2. Answer (3)

Hint & Sol.: 20.96 rounded off to three significant figures will be 21.0

3. Answer (1)

Hint: Dimensional formula of force = $[MLT^{-2}]$

Sol.: $[X] \times [\text{Momentum}] = [\text{Force}]$

$[X]\,[MLT^{-1}] = [MLT^{-2}]$

$[X] = [T^{-1}]$

4. Answer (4)

Hint & Sol.:

$$\text{Plane angle } (\theta) = \left(\frac{\text{Arc}}{\text{Radius}}\right)$$

$$= \frac{[M^0LT^0]}{[M^0LT^0]}$$

$= [M^0L^0T^0]$

5. Answer (2)

Hint: $\Delta L = \Delta L_1 + 2\Delta L_2$

$= 0.01 + 2 \times 0.01$

$= 0.03$

Sol.: $L = L_1 + 2L_2 = 2.02 + 2(1.02)$

$= 4.06$

$L = (4.06 \pm 0.03)$ m

6. Answer (3)

Hint: Use principle of homogeneity

Sol.: $F = at + bt^2$

$[at] = [MLT^{-2}]$

$[a] = [MLT^{-3}]$

$[bt^2] = [MLT^{-2}]$

$[b] = [MLT^{-4}]$

$[a \times b] = [M^2L^2T^{-7}]$

7. Answer (4)

Hint: For a sphere, $A = 4\pi r^2$

$$\frac{\Delta A}{A} \times 100 = 2 \times \frac{\Delta r}{r} \times 100$$

Sol.: $\frac{\Delta A}{A} \times 100 = 2 \times \left(\frac{\Delta r}{r}\right) \times 100$

$$= 2 \times \frac{0.2}{20} \times 100$$

$= 2\%$

8. Answer (1)

Hint: If the number is less than one, then zeroes on the left of first non-zero digit will be insignificant

Sol.: Significant figures in 23.023 m are 5 while significant figures in 0.0003 m are 1

9. Answer (2)

Hint: Use $S = ut + \frac{1}{2}at^2$

Sol.: $30 = 2u + 2a$

$u + a = 15$ ...(i)

$70 = 4u + 8a$ ...(ii)

$70 = 4(15 - a) + 8a$

$a = 2.5$ m/s^2

10. Answer (4)

Hint and Sol.: For a uniform motion acceleration of the body is zero i.e body moves with uniform velocity

11. Answer (2)

Hint: Rocket initially moves with upward acceleration and then it moves under gravity.

Sol.: When acceleration $a = 5$ m/s^2

$t = 120$ s

$$s = ut + \frac{1}{2}at^2$$

$$s_1 = \frac{1}{2} \times 5 \times (120)^2$$

$s_1 = 60 \times 5 \times 120$

$s_1 = 300 \times 120$

$= 36$ km

$v = u + at$

$v = 5 \times 120$

$= 600$ m/s

For free fall motion,

$$s_2 = \frac{v^2}{2g}$$

$$= \frac{(600)^2}{2 \times 10}$$

$$= \frac{600 \times 600}{20}$$

$= 600 \times 30$

$= 18$ km

Total height reached

$H = s_1 + s_2$

$= 36$ km $+ 18$ km

$= 54$ km

12. Answer (2)

Hint: Magnitude of area under velocity-time graph gives distance.

Sol.: $s = 10 + 20 + 10 + 5$

$= 45$ m

13. Answer (1)

Hint: Average speed $= \dfrac{\text{Total distance}}{\text{Total time taken}}$

Sol.: $v_{avg} = \dfrac{x}{\dfrac{2x}{3v_1} + \dfrac{x}{3v_2}}$

$$= \left(\frac{3v_1v_2}{2v_2 + v_1}\right)$$

14. Answer (4)

Hint: Use, $a = \dfrac{dv}{dt}$

Sol.: $a = 4t + 2$

$dv = adt$

$$\int_0^v dv = \int_0^2 (4t + 2)dt$$

$v = [2t^2 + 2t]_0^2$

$= 8 + 4 = 12$ m/s

15. Answer (2)

Hint & Sol.: Distance $= 10 + 20 = 30$ m

Displacement $= -10 - 0 = -10$ m

16. Answer (3)

Hint & Sol.: For a particle moving in a straight line, if acceleration and velocity are in opposite direction, then particle is slowing down. If velocity of particle is zero, then acceleration may be zero or may not be zero.

17. Answer (2)

Hint & Sol.: $R^2 = A^2 + B^2 - 2AB\cos\theta$

For $R = R_{max}$

$\cos\theta = -1$

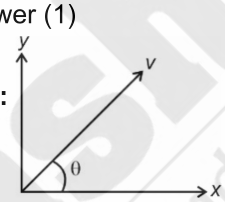
$R = A + B$

18. Answer (1)

Hint & Sol.: For vertically upward projected body angle between acceleration and velocity is 180º, for uniform circular motion angle between acceleration and velocity is 90º, for oblique projectile motion angle between acceleration and velocity changes point to point and for freely falling body angle between acceleration and velocity is zero.

19. Answer (1)

Hint:



$v_x = v\cos\theta$

$v_y = v\sin\theta$

Sol.: $v_x = v\cos\theta$

$24 = 30\cos\theta$

$$\cos\theta = \frac{4}{5}$$

$\theta = 37°$

$v_y = v\sin\theta$

$= 30\sin 37°$

$$= 30 \times \frac{3}{5}$$

$= 18$ m/s

20. Answer (3)

Hint: Use, $H = \dfrac{u^2\sin^2\theta}{2g}$

Sol.: $H = \dfrac{(20)^2\sin^2 30°}{20}$

$$= 20 \times \frac{1}{4}$$

$= 5$ m

Height from the ground $= 60 + 5 = 65$ m

21. Answer (1)

Hint: Use $v = u + at$

Sol.: $u_y = v\sin\theta$

= 30sin30°

= 15 m/s

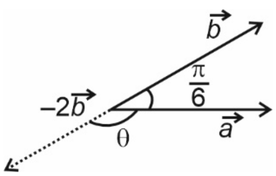
After 1 second,

$v = 15 - 10 \times 1$

= 5 m/s

22. Answer (4)

Hint & Sol.:



$$\theta = \pi - \frac{\pi}{6}$$

$$= \frac{5\pi}{6}$$

23. Answer (2)

Hint: $\vec{v} = \vec{u} + \vec{a}t$

Sol.: $\vec{v} = \left(4\hat{i} + 2\hat{j}\right) + 10\left(0.4\hat{i} + 0.4\hat{j}\right)$

$$\vec{v} = \left(8\hat{i} + 6\hat{j}\right) \text{ m s}^{-1}$$

$$|\vec{v}| = \sqrt{8^2 + 6^2}$$

= 10 m s^{-1}

24. Answer (4)

Hint & Sol.: Two vectors of equal magnitude can have zero resultant in a plane if angle between them is 180°.

25. Answer (4)

Hint: Magnitude of final velocity at highest point = $u\cos\theta$

Sol.: Change in magnitude of velocity = ($u\cos\theta - u$)

26. Answer (4)

Hint: $H = \dfrac{u^2 \sin 2\theta}{2g}$

Sol.: $T = \dfrac{2u\sin\theta}{g}$

$$u\sin\theta = \left(\frac{gT}{2}\right) \; ; \; H = \frac{u^2\sin^2\theta}{2g}$$

$$H = \frac{gT^2}{8}$$

$$\frac{H_1}{H_2} = \frac{T^2}{4T^2}$$

$$H_2 = 4H_1$$

27. Answer (3)

Hint & Sol.: Displacement is a vector quantity which depends only on initial and final position. It may be positive, negative or zero and its magnitude may be less or equal to path length. Path length is scalar quantity which is actual path followed and it is always positive.

28. Answer (2)

Hint: For a given speed, the range of projectile will be equal at angles θ and 90° – θ from horizontal.

Sol.: $T_1 = \dfrac{2u\sin\theta}{g}$

$$T_2 = \frac{2u\cos\theta}{g}$$

$$T_1 \times T_2 = \frac{4u^2\sin\theta \times \cos\theta}{g^2}$$

$$= \frac{2R}{g}$$

29. Answer (3)

Hint: & Sol.: In uniform circular motion direction of centripetal acceleration changes continuously. Velocity and acceleration are always perpendicular to each other.

30. Answer (3)

Hint: $R = \sqrt{a^2 + b^2 + 2ab\cos\theta}$

Sol.: $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

$S^2 = P^2 + Q^2 - 2PQ\cos\theta$

$R^2 + S^2 = 2(P^2 + Q^2)$

31. Answer (4)

Hint: Displacement vector = Final position vector – initial position vector

Sol.: $\vec{r} = \left(3t\hat{i} + 2t^2\hat{j} - t^2\hat{k}\right)$

$$\vec{r}_i = \left(3 \times 1\hat{i} + 2 \times 1\hat{j} - 1\hat{k}\right)$$

$$= \left(3\hat{i} + 2\hat{j} - \hat{k}\right) \text{ m}$$

$$\vec{r}_f = \left(3 \times 2\hat{i} + 2 \times 2\hat{j} - 2^2\hat{k}\right) \text{ m}$$

$= \left(6\hat{i}+8\hat{j}-4\hat{k}\right)$

$\vec{d} = \vec{r}_f - \vec{r}_i$

$= \left(6\hat{i}+8\hat{j}-4\hat{k}\right)-\left(3\hat{i}+2\hat{j}-\hat{k}\right)$

$= \left(3\hat{i}+6\hat{j}-3\hat{k}\right)$ m

32. Answer (2)

Hint & Sol. Work and energy have same dimensional formula [ML^2T^{-2}].

33. Answer (2)

Hint: Use $s = ut + \frac{1}{2}at^2$

Sol.: In *x*-direction

$s = ut + \frac{1}{2}at^2$

$16 = \frac{1}{2}\times 8 \times t^2$

$t = 2$ s

In *y*-direction

$s = ut + \frac{1}{2}at^2$

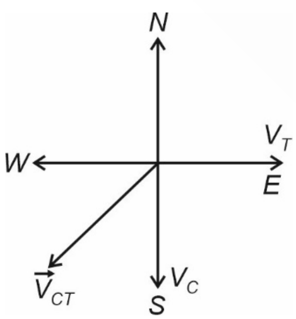
$y = \frac{1}{2}\times 2\times(2)^2 + 10\times 2$

= 24 m

34. Answer (2)

Hint: $\vec{v}_{CT} = \vec{v}_C - \vec{v}_T$

Sol.: $\vec{v}_{CT} = \vec{v}_C - \vec{v}_T$



Velocity of car with respect to train is in South-West.

35. Answer (1)

Hint: Use, $R = \frac{u^2\sin(2\theta)}{g}$

Sol.: $R = \frac{(30)^2\sin 90^\circ}{10}$

= 90 m

Decrease in range due to air resistance

= 90 – 50

= 40 m

SECTION-B

36. Answer (3)

Hint: Drift = time of crossing × speed of river.

Sol.: $t = \frac{d}{V_{SR}}$

$= \frac{100}{5}$

= 20 s

Drift = 20 × 4 = 80 m

37. Answer (1)

Hint: Use, $\tan\theta = \frac{u_y}{u_x}$

Sol.: $u_x = 3$

$u_y = 4$

$\tan\theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$

38. Answer (3)

Hint & Sol. Power $= \frac{\text{Work}}{\text{Time}}$

Power = $\left[\frac{ML^2T^{-2}}{T}\right]$

= [ML^2T^{-3}]

39. Answer (1)

Hint: *Pt* and *qx* are dimensionless.

Sol.: *Pt* = dimensionless

[P] = [T^{-1}]

qx = dimensionless

[q] = [L^{-1}]

[A] = [L]

$\left[\frac{AP}{q}\right] = \left[\frac{LT^{-1}}{L^{-1}}\right]$

= [L^2T^{-1}]

40. Answer (3)

Hint & Sol.: Dimensional formula can be used to check one system of unit to another, to check correctness of equation and to define our own system of unit.

41. Answer (3)

Hint: Use $v^2 = u^2 + 2as$

Sol.: Maximum height $h = \frac{u^2}{2g}$

$v^2 = u^2 + 2as$

$$v^2 = u^2 - 2g \times \frac{3}{4}h$$

$$v^2 = u^2 - 2g \times \frac{3}{4} \times \frac{u^2}{2g}$$

$$v^2 = \frac{u^2}{4}$$

$$v = \frac{u}{2}$$

42. Answer (4)

Hint: Area under acceleration-time graph is equal to change in velocity

Sol.: $\Delta v = \frac{1}{2} \times 8 \times 6$

$v - 0 = 24$ m/s

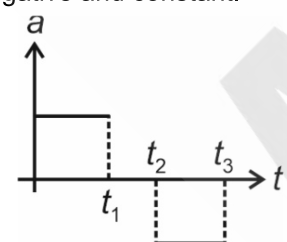
$v = 24$ m/s

43. Answer (2)

Hint: Slope of velocity-time graph is acceleration

Sol.: For time interval 0 to t_1 slope is positive and constant

For time t_1 to t_2 slope is zero and for time t_2 to t_3 slope is negative and constant.



44. Answer (1)

Hint: Use $\hat{n} = \frac{\vec{A} + \vec{B}}{|\vec{A} + \vec{B}|}$

Sol.: $\vec{A} + \vec{B} = 2\hat{i} - \hat{j} + 2\hat{k} + \hat{i} + 2\hat{j} - \hat{k}$

$= 3\hat{i} + \hat{j} + 2\hat{k}$

$|\vec{A} + \vec{B}| = \sqrt{9+1+4} = \sqrt{14}$

$$\frac{\vec{A} + \vec{B}}{|\vec{A} + \vec{B}|} = \left(\frac{3\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{14}}\right)$$

45. Answer (4)

Hint: $a_x = \frac{d^2x}{dt^2}, a_y = \frac{d^2y}{dt^2}$

Sol.: $v_x = \frac{dx}{dt} = \alpha$

$$a_x = \frac{dv_x}{dt} = 0$$

Now, $y = \beta t$

$$v_y = \frac{dy}{dt} = \beta$$

$$a_y = \frac{dv_y}{dt} = 0$$

$a_y = 0$

Hence, a_{net} = zero

46. Answer (2)

Hint: Use, $a_c = \frac{v^2}{R}$

Sol.: $a_c = \frac{(6)^2}{3} = \frac{36}{3} = 12\text{ m/s}^2$

47. Answer (2)

Hint: Time of flight, $T = \sqrt{\frac{2h}{g}}$

Sol.: $2 = \sqrt{\frac{2h}{10}}$

$$4 = \frac{2h}{10}$$

$h = 20$

$u = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = 20$ m/s

Here velocity of projection should be equal to vertical component of velocity.

Hence, $v_{projection} = 20$ m/s

48. Answer (3)

Hint & Sol.: 42 × 0.041 = 1.722

The final significant figures are 2

$\therefore$ Required value is 1.7

49. Answer (4)

Hint: Area under the velocity time graph gives displacement.

Sol.: Displacement

$$= \frac{1}{2}\times 10\times 4+4\times 10+\frac{1}{2}\times 10\times 4=80\text{ m}$$

Average velocity = $\frac{\text{Total displacement}}{\text{Total time taken}}$

$$=\frac{80}{12}=\frac{20}{3}\text{ m/s}$$

50. Answer (1)

Hint: Horizontal velocity of the particle will remains same

Sol.: Horizontal distance

= horizontal velocity × time

= 20 × 2 = 40 m

[CHEMISTRY]

SECTION-A

51. Answer (2)

Hint: Number of atoms = Number of moles × N_A × atomicity

Sol.: • 10 g Ca = $\frac{10}{40}$ × N_A atoms of Ca = $\frac{N_A}{4}$

= $0.25N_A$ atoms

• 10 g N_2 = $\frac{10}{28}$ × N_A × 2 atoms of N = $\frac{5N_A}{7}$ = $0.7N_A$ atoms

• 10 g Al = $\frac{10}{27}$ × N_A atoms of Al = $0.37N_A$ atoms

• 10 g Na = $\frac{10}{23}$ × N_A atoms of Na = $0.43N_A$ atoms

52. Answer (1)

Hint:

$$HNO_3(aq)+NaOH(aq)\longrightarrow H_2O+NO_3^-(aq)+Na^+(aq)$$

Sol.: Number of millimoles of H^+ ions = 100 × 0.2

= 20

Number of millimoles of OH^- ions = 400 × 0.1

= 40

Therefore, number of millimoles of OH^- ions left

= 20

So, molarity of OH^- ions in the solution

$$=\frac{40-20}{400+100}=\frac{20}{500}=0.04\text{ M}$$

53. Answer (4)

Hint: Limestone is $CaCO_3$ and on heating it releases CO_2 gas.

$$CaCO_3(s)\longrightarrow CaO(s)+CO_2(g)$$

Sol.: Mass of pure $CaCO_3$ = (200 × 0.8) g

= 160 g

Since, 100 g of $CaCO_3$ release 22.4 L of CO_2 gas on heating

So, 160 g of $CaCO_3$ on heating releases CO_2 = $\frac{22.4}{100}\times 160$ L

= 35.84 L

54. Answer (2)

Hint: Mass of one He atom = 4 u

Sol.: • 1 amu(u) = 1.66 × 10^{-24} g

• In case of ionic compounds like NaCl, KCl etc. Formula mass is used instead of molecular mass as they do not contain discrete molecules.

55. Answer (3)

Hint: Molality 'm' = $\frac{1000\text{ M}}{1000\text{ d}-MM_1}$ (M_1-molar mass of the solute)

Sol.: Molality 'm' = $\frac{1000}{1500-75}=\frac{1000}{1425}=0.7$ m

56. Answer (2)

Hint & Sol.:

$$C_2H_6(g)+\frac{7}{2}O_2(g)\longrightarrow 2CO_2(g)+3H_2O(l)$$

30 g of C_2H_6 requires $O_2=\frac{7}{2}\times 22.4\text{L}$ at STP

= 78.4 L at STP

So, volume of O_2 required for complete combustion of 7.5 g C_2H_6 = $\frac{78.4}{30}$ × 7.5 L at STP

= 19.6 L

57. Answer (4)

Hint: $A+2B\longrightarrow 3C$

1 mole of A reacts with 2 mole of B to give 3 mole of C.

Sol.: For $A+2B\longrightarrow 3C$

Initial moles 0.8 1.2

0.8 mole of A will react with 1.6 mole of B.

So, B is limiting reagent

1.2 mole of B produces $C = 1.2 \times \frac{3}{2} = 1.8$ mole

58. Answer (1)

Hint: Molality,

$$m = \frac{\text{Number of moles of solute}}{\text{Mass of solvent (in kg)}} = \frac{n}{W}$$

Sol.: $1.5 = \frac{0.3}{W\text{ (in kg)}}$

So, W (in kg) = 0.2 kg = 200 g solvent

59. Answer (4)

Hint: g-atoms or g molecules also represents mol of a substance.

Sol.: • 0.1 mol Ca = 0.1 × 40 = 4 g Ca

• 4.48 L CH_4 gas at STP = 3.2 g CH_4

• 6.022 × 10^{22} atoms of Na = 0.1 mol Na = 2.3 g Na

• 0.1 g atom Cu = 0.1 mol Cu = 6.35 g Cu

60. Answer (2)

Hint: In a compound atoms of different elements have a fixed ratio of their masses.

Sol.:

Element	Percentage composition	Atomic mass	Relative number of atoms	Ratio of atoms
C	40	12	$\frac{40}{12} = 3.33$	1
H	6.66	1	$\frac{6.66}{1} = 6.66$	2
O	53.33	16	$\frac{53.33}{16} = 3.33$	1

Empirical formula of the compound is CH_2O

61. Answer (2)

Hint: Radius of n^{th} orbit in H-like species is given as $r_n = 0.529\frac{n^2}{Z}$ Å

Sol.: $\frac{r_{He^+(n=1)}}{r_{Li^{2+}}(n=2)} = \frac{\frac{1}{2}}{\frac{4}{3}} = \frac{3}{8}$

$\therefore r_{Li^{2+}} = \frac{8}{3} r_{He^+} = \frac{8x}{3}$

62. Answer (3)

Hint: Chemical species having equal number of electrons are called isoelectronic species.

Sol.: CO, N_2, NO^+ and CN^- all have 14 electrons each.

63. Answer (1)

Hint: Size and energy of orbitals $\propto$ principal quantum number (n)

Sol.: As principal quantum number increases energy and size of a particular type of orbital increases.

So, size and energy of orbitals decreases as 4p > 3p > 2p.

64. Answer (4)

Hint: Wavelength of photon emitted $\propto \frac{1}{Z^2}$; 'Z' is the atomic number.

Sol.: Be^{3+} ; Z = 4

$\therefore$ Transition (n = 3 to n = 2) will have shortest wavelength.

65. Answer (3)

Hint: $P + 3\bar{e} \rightarrow P^{3-}$

Sol.: Number of electrons in P^{3-} ion = 15 + 3 = 18

Number of protons in P^{3-} ion = 15

Number of neutrons in P^{3-} ion = 31 – 15 = 16

66. Answer (1)

Hint: The square of the wave function (i.e. Ψ^2) at a point gives the probability density of the electron at that point.

Sol.: For 1*s* orbital, probability density is maximum at nucleus and decreases sharply as we move away from it.

At nodes, $|\Psi^2|$ is zero.

67. Answer (2)

Hint: Orbital angular momentum is given as

$$m_\ell = \sqrt{\ell(\ell+1)}\frac{h}{2\pi}$$

Sol.: For a d electron, ℓ = 2

$\therefore$ Angular momentum = $\sqrt{2(2+1)}\frac{h}{2\pi} = \sqrt{6}\frac{h}{2\pi}$

68. Answer (1)

Hint: For H atom energy of orbitals depend on 'n' value only.

Sol.: Order of energy of orbitals, for (H) atom $1s < 2s = 2p < 3s = 3p = 3d$

For multielectron system, more the value of (n + l), higher will be the energy

Orbital	**n + R**
$3d$	$s + 2 = 5$
$4s$	$4 + 0 = 4$

69. Answer (3)

Hint: Number of radial nodes for an orbital = $n - \ell - 1$

Sol.: Number of radial nodes = $n - \ell - 1$

So, for 3p orbital, radial nodes = 3 – 1 – 1 = 1

Number of angular nodes = ℓ

So, for 3p orbital, angular nodes = 1

70. Answer (2)

Hint: Diffraction and interference are explained by the wave nature of electromagnetic radiation

Sol.: Black body radiation and photoelectric effect are explained by particle nature of electromagnetic radiation.

71. Answer (3)

Hint: $m_s = +\frac{1}{2}$ represents an electron in an orbital having clockwise spin.

Sol.: For n = 4, there could be maximum $2n^2 = 32$ electrons, out of that maximum 16 electrons can have their spin clockwise i.e., $m_s = +\frac{1}{2}$.

72. Answer (2)

Hint: Number of waves formed in n^{th} orbit = n

Sol.: The number of waves formed in the fourth orbit of H atom is four.

73. Answer (3)

Hint: Two electrons from 4s subshell are removed to convert Ni to Ni^{2+} ion.

Sol.: E.C. of Ni(28) : $[Ar]3d^8, 4s^2$

So, E.C. of Ni^{2+} ion is $[Ar]3d^8$

Fe(26) : $[Ar]3d^6, 4s^2$

So E.C. of Fe^{3+} ion is $[Ar]3d^5$

V(23) = $[Ar]3d^3 4s^2$

V^{3+} = $[Ar]\ 3d^2$

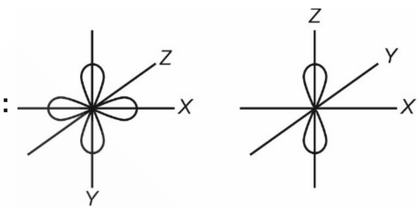
Co(27) = $[Ar]\ 3d^7 4s^2$

Co^{2+} = $[Ar]\ 3d^7$

74. Answer (1)

Hint: An orbital can have maximum two electrons having opposite spins.

Sol.:



$d_{x^2-y^2}$ and p_z orbitals have electron density along the axes.

75. Answer (4)

Hint: Molar mass of NH_3 and CO_2 are 17 g and 44 g respectively.

Sol.:

$$\frac{\text{Mass of } NH_3}{\text{Mass of } CO_2} = \frac{\text{Number of moles} \times 17}{\text{Number of moles} \times 44} = \frac{2 \times 17}{1 \times 44}$$

= 17 : 22

76. Answer (1)

Hint: Number of electrons present in Mg^{2+} ion = 10

Sol.: Number of moles of Mg^{2+} ions $= \frac{2.4}{24} = \frac{1}{10}$

Number of Mg^{2+} ions $= \frac{N_A}{10}$

So, total number of electrons in 2.4 g Mg^{2+} ions $= \frac{N_A}{10} \times 10 = N_A$

77. Answer (3)

Hint: Shape of an orbital is determined by azimuthal quantum number 'l'.

Sol.: • Be^{2+} has 2 electrons while He^+ has only one electron.

• In n^{th} shell, total number of orbitals = n^2

78. Answer (4)

Hint: 'l' values vary from 0 to (n – 1)

m values vary from –l to +l

Sol.: For l = 1, m = –1, 0, +1

For n = 1, l = 0

79. Answer (3)

Hint: Moving down the group work function of metal decreases as effective nuclear charge decreases.

Sol.: Work function of Na > work function of Cs.

There is no time lag between the striking of light beam and ejection of photoelectron from metal surface.

80. Answer (1)

Hint & Sol.: Correct decreasing order of frequencies of radiations is as

X-rays > UV > IR > Radio waves

81. Answer (2)

Hint: For polyatomic gases, number of atoms = Number of moles of gas × atomicity of the gas

Sol.: • 8 g O_2 = $\frac{8}{32} \times 2$ N_A atoms of O

$= \frac{N_A}{2}$ atoms of O

• 9 g H_2O i.e. $\frac{N_A}{2}$ molecules of H_2O also contains $\frac{N_A}{2}$ atoms of O.

82. Answer (3)

Hint: 2 × vapour density = Molar mass

Sol.: Molar mass = 2 × 23 = 46

So, possible gas is NO_2

83. Answer (2)

Hint: Number of moles = $\frac{\text{Given mass (in g)}}{\text{Molar mass}} = \frac{N}{N_A}$

Sol.: Number of molecules left = $\left(\frac{220 \times 10^{-3}}{44} \times 6.022 \times 10^{23}\right) - 10^{21}$

= $3.011 \times 10^{21} - 10^{21}$

= 2.011×10^{21}

84. Answer (3)

Hint: $CO_2 + C \xrightarrow{\Delta} 2CO$

Sol.: According to the reaction of CO_2 with hot coke

$$\underset{1\,\text{mole} = 44\text{ g}}{CO_2} + C \xrightarrow{\Delta} \underset{56\text{ g}}{2CO}$$

Increase in mass of gas = 56 – 44 = 12 g

So, percentage increase in mass = $\frac{12}{44} \times 100$

= 27.27%

85. Answer (3)

Hint: The energy gap between the two orbits is given by equation $\Delta E = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ where $n_2 > n_1$

Sol.: Energy of the photon emitted during electronic transition from n = 5 to n = 1 is gives as

$$\Delta E = R_H\left(\frac{1}{1^2} - \frac{1}{5^2}\right) = R_H\left(\frac{24}{25}\right)$$

- ΔE for transition of electron from n = 2 to n = 1 is

$$\Delta E = R_H\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R_H\left(\frac{3}{4}\right)$$

SECTION-B

86. Answer (2)

Hint: Mass of one water molecule = 18 u

= $18 \times 1.67 \times 10^{-24}$ g

Sol.: Volume = $\frac{\text{Mass(g)}}{\text{Density }(\text{gcm}^{-3})}$

So, volume of one H_2O molecule

$= \frac{18 \times 1.67 \times 10^{-24}\text{ g}}{1\text{ g cm}^{-3}}$

= 3.02×10^{-23} cm^3

• 5.6 L of SO_2 gas at STP = $\frac{1}{4}$ mol of SO_2 gas

It contains $\frac{1}{4}$ mol of sulphur i.e. 8 g of sulphur.

87. Answer (4)

Hint: Zero between two non-zero digits are significant

Sol.: 2.005 has 4 significant figures.

88. Answer (3)

Hint: The filling of electrons into the orbitals of different atoms takes place according to the Aufbau principle, Hund's rule of maximum multiplicity and Pauli's exclusion principle.

Sol.: According to Hund's rule of maximum multiplicity "pairing of electrons in the orbitals belonging to the same subshell does not take place until each orbital belonging to that subshell is singly occupied".

89. Answer (4)

Hint: Pd (Z = 46) has exceptional electronic configuration as it does not follow Aufbau principle.

- There are total n subshells in the n^{th} shell.

Sol.: • E.C. of Pd (Z = 46) : [Kr] $4d^{10}$

- Two electrons occupying the same orbital have different values of spin quantum number and value.

90. Answer (4)

Hint: For the *h*-subshell, azimuthal quantum number l = 5 and values of l varies from 0 to (n – 1)

Sol.: For a shell having *h* subshell (l = 5) lowest value of n should be 6 as for n = 6, l = 0, 1, 2, 3, 4, 5

91. Answer (1)

Hint: E.C. of Cr^{3+} is $[Ar]3d^3$, so it has 3 unpaired electrons

Sol.:

Cr^{3+}	$[Ar]3d^3$	3
Mn^{2+}	$[Ar]3d^5$	5
Fe^{2+}	$[Ar]3d^6$	4
Ni^{2+}	$[Ar]3d^8$	2

92. Answer (4)

Hint: Total number of orbitals for a given subshell(l) is equal to (2l + 1)

Sol.: For a given subshell 'l' since number of orbitals = 2l + 1

So, number electrons in a given subshell = 2(2l + 1)

= 4l + 2

93. Answer (2)

Hint: Isobars have same atomic mass but different atomic numbers.

Sol.: ${}^{35}_{17}Cl$ and ${}^{37}_{17}Cl$: Isotopes

${}^{15}_{7}N$ and ${}^{16}_{8}O$: Isotones

${}^{30}_{14}Si$ and ${}^{31}_{15}P$: Isotones

Ar and F_2 : Isoelectronic

94. Answer (3)

Hint: According to Rutherford's atomic model

$r_{atom} = 10^{-10}$ m

$r_{nucleus} = 10^{-15}$ m

Sol.: Heisenberg's uncertainty principle is significant only for motion of microscopic objects. According to de-Broglie, every object in motion has a wave character.

95. Answer (2)

Hint: For an electron moving in an orbit (n) with radius (r), de Broglie wavelength(λ) is related as $2\pi r = n\lambda$

Sol.: For H-atom radius of 1st orbit = x then the radius of 2^{nd} orbit of He^+ ion = $x \times \frac{2^2}{2}$

$= 2x$

Now, using the relationship

$2\pi r = n\lambda$

$2\pi(2x) = 2\lambda$

$\therefore \quad \lambda = 2\pi x$

96. Answer (4)

Hint: NaCl and is 1:1 type salt

Sol.: Number of millimoles of NaCl = 0.1 × 500 = 50

So, number of millimoles of Na^+ and Cl^- ions = 50

Number of millimoles $Ba(NO_3)_2$ = 0.2 × 1500 = 300

So, number of millimoles of Ba^{2+} and NO_3^- ions = 300 and 600

Now, therefore concentration of Na^+ ion ≡ $[Na^+]$ =

$\frac{50}{2000} = \frac{1}{40}$M

$\therefore \quad [Cl^-] = \frac{1}{40}$M

Also $[Ba^{2+}] = \frac{300}{2000} = \frac{3}{20}$ M

and $[NO_3^-] = \frac{600}{2000} = \frac{3}{10}$M

97. Answer (3)

Hint: An element may have different isotopes with unequal proportions.

Sol.: In calculations, average atomic mass of various isotopes of an element is taken.

98. Answer (4)

Hint: $CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + H_2O + CO_2$

Sol.: Number of moles of H_2SO_4 = Number of moles of $CaCO_3$

$= \frac{10}{100} = \frac{1}{10}$

So, mass of $H_2SO_4 = \frac{1}{10} \times 98\text{ g} = \frac{49}{5}\text{g}$

Therefore mass of solutions

$= \frac{49}{5} \times \frac{100}{60}\text{g}$

= 16.33 g

99. Answer (3)

Hint: One mole of $Ba_3(PO_4)_2$ contains 3 mole of Ba, 2 mole of P and 8 mole of O.

Sol.: In 1 mole of $Ba_3(PO_4)_2$ there is 8 moles of O = 128 g of oxygen, so 0.1 mole of $Ba_3(PO_4)_2$ contains 12.8 g of oxygen.

100. Answer (1)

Hint: For shortest wavelength energy difference should be maximum.

Sol.: $\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{\infty}\right] = \frac{1}{\lambda_1}$

So, $R_H = \frac{1}{\lambda_1}$

Now for He⁺ ion $\frac{1}{\lambda_{He^+}} = R_H(2)^2\left[\frac{1}{9} - \frac{1}{16}\right]$

$= R_H \frac{7}{36}$

$\therefore \quad \lambda_{He^+} = \frac{36}{7}\lambda_1$

[BOTANY]

SECTION-A

101. Answer (2)

Hint: Endoplasmic reticulum is involved in synthesis of proteins.

Sol.: Golgi apparatus is the important site of formation of glycoproteins and glycolipids.

102. Answer (3)

Hint: Ribosomes are membraneless organelles involved in protein synthesis.

Sol.: Nucleolus is a site for active ribosomal RNA synthesis.

103. Answer (3)

Hint: Meiosis involves two sequential cycles of nuclear and cell division.

Sol.: Tetrads are more clearly visible in pachytene.

104. Answer (3)

Hint: Interphase is called the resting phase of cell cycle.

Sol.: M-phase is the phase where actual cell division occurs. In this phase, cell utilizes the constituents formed during interphase and divide into two daughter cells.

105. Answer (4)

Hint: In the end stage of karyokinesis, the chromosomes reach their respective poles.

Sol.: Anaphase of mitosis is the best stage to study the shape of chromosomes.

106. Answer (4)

Hint: Centriole gives a cartwheel appearance.

Sol.: Bacteria also possess flagella but these are structurally different from that of the eukaryotic flagella. Centrioles help in the formation of basal bodies which give rise to cilia and flagella.

107. Answer (3)

Hint: Tribe is an intermediate category between sub-family and genus.

Sol.: Species serves as the basic and lowest category in a taxonomic hierarchy.

108. Answer (4)

Hint: Brinjal belongs to the family Solanaceae.

Sol.: Polymoniales is the order of brinjal.

109. Answer (3)

Hint: Crossing over takes place in the third substage of prophase I.

Sol.: The correct sequence of events during prophase I of meiosis are as follows:

Condensation and coiling of chromatin fibres begins → Formation of synaptonemal complex → Appearance of recombination nodules → Appearance of X-shaped structures → Spindle assembles to prepare homologous chromosomes for separation.

110. Answer (3)

Hint: Amyloplast is starch containing leucoplast.

Sol.: Elaioplasts store fats and oils.

111. Answer (4)

Hint: Family is a group of related genera.

Sol.: A class is a group of one or more related orders.

112. Answer (2)

Hint: Function of mitochondria is not coordinated with endomembrane system.

Sol.: The stroma of chloroplast contains enzymes required for carbohydrates and proteins synthesis.

The hydrolytic enzymes of lysosomes are active at acidic pH. The ER, Golgi apparatus, lysosomes and vacuoles are included in endomembrane system.

113. Answer (4)

Hint: Pachytene is characterised by appearance of recombination nodules.

Sol.: Diplotene can last for months or years in the oocytes of vertebrates.

114. Answer (2)

Hint: The phenomenon of bringing the chromosomes on the equator of spindle is called congression.

Sol.: Congression is observed in metaphase.

115. Answer (1)

Hint: Telocentric chromosome has a terminal centromere.

Sol.: In the acrocentric chromosome, one arm is extremely short and the other is very long. *Cis* and *trans* face of golgi apparatus are different but interconnected.

116. Answer (2)

Hint: Mitochondria are the powerhouses of the cell.

Sol.: Golgi apparatus is involved in the formation of lysosomes. Mitochondrion is the site of aerobic respiration. Contractile vacuole is important for osmoregulation and excretion. SER is involved in detoxification of drugs.

117. Answer (3)

Hint: G_2 phase is the second gap phase present between S and M phase.

Sol.: DNA synthesis stops at this phase but cell synthesizes RNA, proteins etc. required during the next phase.

118. Answer (2)

Hint: Meiosis I occurs in diploid cells.

Sol.: Anaphase II occurs in haploid cells.

119. Answer (4)

Hint: Cell plate formation occurs centrifugally.

Sol.: Terminalisation of chiasmata occurs in diakinesis stage. When karyokinesis is not followed by cytokinesis, formation of syncytium takes place.

120. Answer (4)

Hint: Biodiversity refers to number and types of organisms present on earth.

Sol.: The number of species that are known and described range between 1.7 – 1.8 million.

121. Answer (3)

Hint: Aristotle divided living beings into animals and plants.

Sol.: Carolus Linnaeus developed the binomial nomenclature for scientific naming of organisms.

122. Answer (4)

Hint: *Cis* face is the forming face of Golgi.

Sol.: Material to be packaged in the form of vesicles from ER fuses with *cis* face of Golgi.

123. Answer (4)

Hint: The amount of lipids is lesser than proteins in RBC membrane of humans.

Sol.: In human erythrocyte, membrane has approximately 52% proteins and 40% lipids.

124. Answer (3)

Hint: Schleiden and Schwann together formulated the cell theory.

Sol.: Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells.

125. Answer (4)

Hint: New mitochondria and chloroplasts arise by the division of pre-existing ones.

Sol.: Both have their own nucleic acids (circular ds DNA and RNA) and 70S ribosomes. Their membranes resemble with that of bacteria.

126. Answer (4)

Hint: Taxonomy deals with the study of principles and procedures of classification.

Sol.: Taxonomy includes identification, nomenclature and characterisation.

127. Answer (2)

Hint: Plant taxonomists have evolved International Code of Botanical Nomenclature.

Sol.: ICNB stands for International Code of Nomenclature of Bacteria.

128. Answer (2)

Hint: Mesosomes are the structures that are formed by invagination of plasma membrane into the cell.

Sol.: Mesosome is found in Gram positive bacteria.

129. Answer (4)

Hint: Glycogen granules do not have a membrane around them.

Sol.: Vacuoles are membrane bound structures.

130. Answer (3)

Hint: The shape of cell may vary with the function they perform. *Eg*. Nerve cells are long and branched.

Sol.: Tracheids – Elongated

Mesophyll cells – Round and oval

White blood cells – Amoeboid

Red blood cells – Round and biconcave

131. Answer (4)

Hint: Single mRNA with multiple ribosomes is called polysome.

Sol.: SER is the site of steroidal hormones synthesis. Polysome is not an inclusion body. Chromatophores take part in photosynthesis.

132. Answer (2)

Hint: Synaptonemal complex is formed during second sub-stage of prophase-I.

Sol.: The beginning of diplotene is recognised by the dissolution of synaptonemal complex.

133. Answer (2)

Hint: The first word denoting genus starts with capital letter.

Sol.: Scientific name is printed in italics or underlined separately when handwritten to indicate its Latin origin.

134. Answer (4)

Hint: Meiosis ultimately plays role in evolution.

Sol.: At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between two daughter cells.

135. Answer (2)

Hint: Singer and Nicolson (1972) described the plasma membrane as the protein icebergs floating in the sea of lipids.

Sol.: The fluid nature of plasma membrane is important for the functions like cell growth, formation of intercellular functions, secretion etc.

Vacuole is a membrane-bound space found in the cytoplasm.

SECTION - B

136. Answer (3)

Hint: Plant cells have large vacuoles.

Sol.: In plant cells, centrosomes are absent, vacuole is permanent and large and plastids are present.

137. Answer (4)

Hint: Meiosis is the reductional division in which number of chromosomes is reduced to half.

Sol.: Amount of DNA in meiotic II products = $\frac{40}{4}$

$\Rightarrow$ 10 pg

Number of chromosomes is 8.

138. Answer (3)

Hint: Human cell cycle lasts for about 24 hours.

Sol.: In a human cell cycle, cell division proper lasts for 1 hour.

139. Answer (3)

Hint: G_2 phase is also called as pre-mitotic phase.

Sol.: Most of the organelle duplication occurs in G_1-phase.

140. Answer (1)

Hint: The number of common characteristics goes on decreasing as we go from species to kingdom.

Sol.: Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

141. Answer (4)

Hint: Members of Poaceae family belong to monocotyledonae.

Sol.: Cats – Felidae

Housefly – Diptera

Triticum – Monocotyledonae

Mangifera – Anacardiaceae

142. Answer (1)

Hint: Family is a group of related genera.

Sol.: *Solanum*, *Petunia* and *Datura* belong to the family solanaceae. Cats and dogs have canine teeth and are placed in the same order.

143. Answer (4)

Hint: Metaphase is the best stage to study the morphology of chromosomes.

Sol.: Spireme stage is seen in prophase. Chromosomes are thickest and shortest in metaphase. Centromeres split in anaphase and chromosomes lose their identity as discrete elements in telophase.

144. Answer (3)

Hint: Interkinesis is a metabolic stage between telophase I and prophase II.

Sol.: During interkinesis, no DNA replication occurs. In male honey bees, haploid cells divide by mitosis.

145. Answer (4)

Hint: Based on the internal, external and other characteristics, all living organisms can be classified into different taxa.

Sol.: External and internal structure, along with cell structure, development process and ecological information of organisms are essential and form the basis of modern taxonomic studies.

146. Answer (1)

Hint: Systematics is the wider field of science as with identification, nomenclature and classification, it also takes into account evolutionary relationships.

Sol.: The word systematics is derived from Latin word, *"Systema"*.

147. Answer (3)

Hint: Mitosis produces new cells for healing the wounds and for regeneration.

Sol.: Formation of double metaphasic plate occurs in metaphase I of meiosis I.

148. Answer (1)

Hint: Genus is a group of related species, which has more characters in common in comparison to the species of another genera.

Sol.: *familiaris*, *lupus*, *aureus*, belong to the same genus *Canis*.

149. Answer (2)

Hint: A descending sequence progresses from kingdom to species.

Sol.: The correct descending sequence is as follows:

Animalia (Kingdom) → Chordata (Phylum) → Mammalia (Class) → Primata (Order) → Hominidae (Family)

150. Answer (3)

Hint: A species can be easily distinguished from other closely related species on the basis of their distinct morphological differences.

Sol.: Members of a species can interbreed to produce fertile offsprings. Variations and gene flow can occur between populations of a species.

[ZOOLOGY]

SECTION-A

151. Answer (3)

Hint: A coelenterate

Sol.: In unicellular organisms like *Amoeba*, *Paramoecium* and yeast, all functions like digestion, respiration and reproduction are performed by a single cell. In the body of complex animals, the same basic functions are carried out by different groups of cells in a well organised manner. The body of a simple organism like *Hydra* is made up of different types of cells and the number of cells in each type can be in thousands.

152. Answer (2)

Hint: Simple squamous epithelium

Sol.: The squamous epithelium is made of a single thin layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs.

The cuboidal epithelium is composed of a single layer of cube-like cells.

The columnar epithelium is composed of a single layer of tall and slender cells with basal nuclei.

153. Answer (4)

Hint: Non-nutrient chemical

Sol.: Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products. These products are released through ducts or tubes. In contrast, endocrine glands do not have ducts. Their products called hormones are secreted directly into the fluid bathing the gland.

154. Answer (1)

Hint: Trichloroacetic acid

Sol.: To perform a chemical analysis, a living tissue is grinded in trichloroacetic acid using a mortar and pestle. We obtain a thick slurry which is then strained through a cheesecloth or cotton to obtain two fractions. One is called filtrate and the second is called retentate.

155. Answer (2)

Hint: % weight of nitrogen in human's body is 3.3

Sol.:

Element	**% Weight of Earth's crust**	**% Weight of Human body**
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	Very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	Negligible

156. Answer (1)

Hint: **'X'** has a single 'double bond' in its ring

Sol.: There are four different nitrogenous bases commonly found in DNA ; adenine (A), guanine (G), thymine (T) and cytosine (C). RNA also contains adenine, guanine and cytosine but instead of thymine, it has uracil (U). Adenine and guanine are double ring bases called substituted purines. Cytosine, thymine and uracil are single ring bases called substituted pyrimidines.

Guanine and cytosine show complementary base pairing and form three hydrogen bonds, while adenine and thymine bind with each other by two hydrogen bonds.

157. Answer (2)

Hint: Basic amino acid

Sol.: Basic amino acids have an additional amino group. For *eg* – Lysine, arginine, *etc.*

Neutral amino acids are the ones that possess equal number of amino group and carboxylic group. For *eg*: Valine, glycine, *etc*.

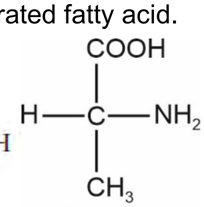
Serine is an alcoholic amino acid and phenylalanine is an example of aromatic amino acid.

158. Answer (3)

Hint: Palmitic acid is a saturated fatty acid.

Sol.: $CH_3-(CH_2)_{14}-COOH$

Fatty acid
(Palmitic acid)



(Alanine)

159. Answer (3)

Hint: Protein

Sol.: Inulin, glycogen and chitin are homopolysaccharides while GLUT-4 and antibody belong to the category of proteins.

GLUT-4 enables glucose transport into the cells while antibody fights infectious agents.

160. Answer (4)

Hint: Belongs to the category of specialised connective tissue

Sol.: Collagen is the most abundant protein in the animal world. Tendons and ligaments belong to the category of dense regular connective tissues in which orientation of collagen fibres show a regular pattern.

Blood is a fluid connective tissue that lacks collagen fibres.

161. Answer (2)

Hint: Compound epithelium

Sol.: Compound epithelium is made of more than one layer of cells and thus has a limited role in secretion and absorption. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

Simple squamous epithelium lines the alveoli of lungs.

Proximal convoluted tubule is lined by the brush bordered cuboidal epithelium.

Ciliated columnar epithelium lines the fallopian tube that aids in transport of ovum.

162. Answer (4)

Hint: Starch is present as the storage polysaccharide in plants.

Sol.: Starch is present as a store house of energy in plant tissues. It forms helical secondary structures. In fact, starch can hold I_2 molecules in the helical portion. The starch-I_2 complex is blue in colour. Cellulose does not contain complex helices and hence, cannot hold I_2.

163. Answer (2)

Hint: Cardiac muscle fibres

Sol.: Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

Cardiac muscle fibres show densely stained cross-bands called intercalated discs having communication junctions at intervals.

Cell junctions hold smooth muscle fibres together and they are bundled together in a connective tissue sheath.

164. Answer (3)

Hint: Similar to ricin

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc*.
Alkaloids	Morphine, Codeine, *etc*
Terpenoides	Monoterpenes, Diterpenes
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

165. Answer (2)

Hint: Adipose tissue

Sol.: Adipose tissue is a type of loose connective tissue which is located mainly beneath the skin. The cells of this tissue are specialised to store fats. The excess of nutrients which are not used immediately are converted into fats and are stored in this tissue. In dense connective tissue, fibres and fibroblasts are compactly packed.

Blood and lymph are examples of fluid connective tissues.

166. Answer (4)

Hint: Include fatty acid

Sol.: Ester bond, glycosidic bond and peptide bond are formed by dehydration reaction (elimination of a water molecule). Insulin, glucagon and receptor are proteinaceous in nature. Cellulose and glycogen are polysaccharides in which the monomeric units *i.e.*, glucose, are joined by glycosidic bonds.

Lecithin is a phospholipid: Nucleotides are also formed by dehydration synthesis forming ester and glycosidic bonds.

167. Answer (3)

Hint: Increases the strength of bones

Sol.: Bones have a hard and non-pliable ground substance, rich in calcium salts and collagen fibres which give bone its strength. The bone cells (osteocytes) are present in the spaces called lacunae.

The intercellular material of cartilage is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them.

168. Answer (1)

Hint: Also present between two nitrogenous bases

Sol.: The long protein chain or the polypeptide chain usually folds upon itself like a hollow woolen ball. This is termed as the tertiary structure. This structure gives a 3-dimensional view of a protein.

Tertiary structure is absolutely necessary for the many biological activities of proteins.

Hydrogen bond, ionic bond, covalent bond, hydrophobic bond as well as van der Waals interaction are involved during coiling of a polypeptide.

169. Answer (4)

Hint: Neuroglia

Sol.: Neural tissue exerts the greatest control over the body's responsiveness to changing conditions. Neurons, the functional units of neural system are excitable cells due to a differential concentration gradient of ions across the membrane.

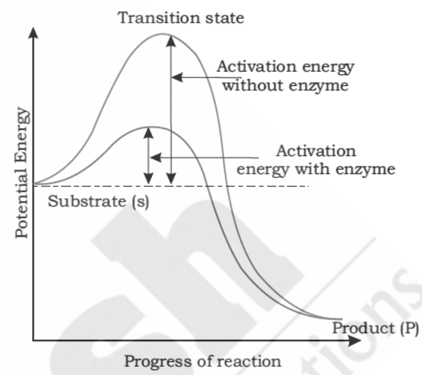
The neuroglial cells which constitute the rest of the neural system protect and support neurons.

170. Answer (2)

Hint: Intermediate structure

Sol.:

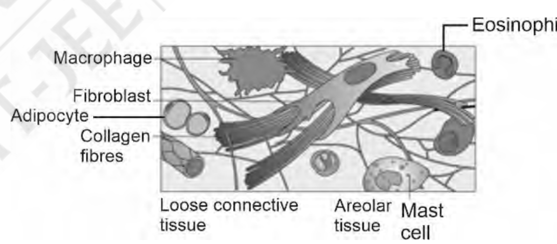
The graph represents an exothermic reaction.



171. Answer (3)

Hint: Possesses the cells that are present in adipose tissue as well

Sol.:



172. Answer (3)

Hint: Hydrolases

Sol.: Hydrolases catalyse the hydrolysis of ester, ether, peptide, glycosidic , C-C, C-halide or P–N bonds.

Hydrolases belong to class III of enzymes.

Oxidoreductases or dehydrogenases belong to class I of enzymes.

Transferases and lyases belong to class II and class IV of enzymes respectively.

173. Answer (4)

Hint: Tubular parts of nephron is lined by simple cuboidal epithelium

Sol.:

Simple squamous epithelium	– Forms diffusion boundary
Simple cuboidal epithelium	– Assists in secretion and absorption
Ciliated epithelium	– Moves mucus in a specific direction
Compound epithelium	– Provides protection against chemical and mechanical stresses

174. Answer (3)

Hint: Enzymes remain in a temporarily inactive state.

Sol.: Enzymes generally function in a narrow range of temperature and pH. Each enzyme shows its highest activity at a particular temperature and pH called the optimum temperature and optimum pH. Activity declines both below and above the optimum value. Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys the enzymatic activity because proteins are denatured by heat.

175. Answer (1)

Hint: Connective tissue

Sol.: Connective tissues are the most abundant and widely distributed tissue in the body of complex animals. They are named connective tissues because of the special function of linking and supporting other tissues/organs of the body. They range from soft connective tissues to specialised types.

Muscles play an active role in all the movements of the body.

Epithelial tissues line the body cavities, ducts and tubes.

176. Answer (3)

Hint: 1 full turn = 360º

Sol.: According to Watson and Crick model, the DNA exists as a double helix. The two strands of polynucleotides are antiparallel *i.e.*, run in the opposite direction.

In B-DNA, at each step of ascent, the strand turns 36º. One full turn of the helical strand (360º) would involve ten base pairs. The base pairs in DNA are stacked 3.4 Å apart. Thus, pitch of the DNA is 34 Å as ten base pairs occupy a distance of about 34 Å.

177. Answer (4)

Hint: Cardiac muscle is present in the heart.

Sol.: The wall of internal organs such as the blood vessels, stomach and intestine contains smooth muscle tissues.

Skeletal muscle fibres are present in the limbs, body wall, face, neck, *etc*.

Skeletal as well as cardiac muscle fibres are striated and cylindrical in shape.

Skeletal as well as smooth muscle fibres do not possess intercalated discs.

Both smooth as well as cardiac muscle fibres are involuntary in action.

178. Answer (3)

Hint: Glycosidic bond

Sol.: Carbohydrates constitute 3% of the total cellular mass. In a polysaccharide, the individual monosaccharides are linked by glycosidic bonds.

Cotton fibres are cellulosic and its monomeric units *i.e.*, glucose, are joined together by glycosidic bonds.

In nucleosides, the nitrogenous base is joined to the sugar molecule by a glycosidic bond.

RuBisCO are the most abundant protein in the whole of the biosphere. The monomeric units of proteins are joined by peptide bonds.

179. Answer (1)

Hint: Plays a crucial role in the catalytic activity

Sol.: Co-enzymes are organic compounds, whose association with the apoenzyme is only transient, usually occurring during the course of catalysis. Furthermore, co-enzymes serve as co-factors in a number of different enzyme catalysed reactions. The essential chemical components of many co-enzymes are vitamins, *e.g.*, co-enzyme Nicotinamide Adenine Dinucleotide (NAD) and NADP contains the vitamin niacin.

Catalytic activity is lost when the co-factor is removed from the enzyme which testifies that they play a crucial role in the catalytic activity of the enzymes.

180. Answer (4)

Hint: Class IV

Sol.: Carbonic anhydrase catalyses the formation of carbonic acid from CO_2 and H_2O.

A four-digit enzyme commission (EC) number is assigned to each enzyme in which

Ist digit $\rightarrow$ Class

IInd digit $\rightarrow$ Sub class

IIIrd digit $\rightarrow$ Sub-sub class

IVth digit $\rightarrow$ Individual enzyme

181. Answer (3)

Hint: Increases or decreases in either direction.

Sol.: A general rule of thumb is that rate doubles or decreases by half for every 10 ºC change in either direction.

- Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get denatured at high temperatures.
- Almost all enzymes are proteins. There are some nucleic acids that behave like enzymes. These are called ribozymes.
- Biomolecules are carbon compounds present in living tissues.

182. Answer (2)

Hint: Columnar and cuboidal cells

Sol.: The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys.

Columnar epithelium are found in the lining of stomach and intestine.

Some of the columnar or cuboidal cells get specialised for secretion and are called glandular epithelium.

183. Answer (1)

Hint: Adenosine triphosphate

Sol.: The most important form of energy currency in living systems is the bond energy in a chemical called adenosine triphosphate (ATP). It possesses three phosphate groups.

- Two hydrogen bonds are present between adenine and thymine.
- Lecithin is a phospholipid which is composed of two fatty acids, a phosphate group and a nitrogen containing choline molecule.
- Glycine is the simplest amino acid.
- Two hydrogen atoms are directly attached to the α-C atom in glycine.

184. Answer (2)

Hint: Haem is the prosthetic group.

Sol.: Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyze the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.

Zn^{2+} is the co-factor for the enzymes, carboxypeptidase and carbonic anhydrase.

185. Answer (2)

Hint: Exclude proteins

Sol.: Primary metabolites have identifiable functions and play known roles in the physiological processes.

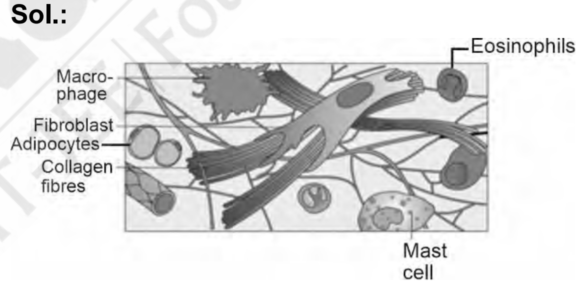
- Cellulose is a secondary metabolite. Glucagon and insulin are hormones that are proteinaecous in nature.
- Inulin is a polymer of fructose.

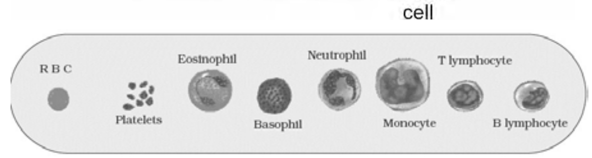
SECTION-B

186. Answer (3)

Hint: Type of granulocyte

Sol.:





Adipocytes are fat storage cells.

Macrophages are distributed in tissues.

187. Answer (4)

Hint: Catabolic pathway

Sol.: Metabolic pathways can lead to a more complex structure from a simpler structure (for example, acetic acid becomes cholesterol) or lead to a simpler structure from a complex structure (for example, glucose becomes lactic acid). The former cases are called biosynthetic pathways or anabolic pathways. The latter constitute degradation and hence are called catabolic pathways. Anabolic pathways, as expected, consume energy. Assembly of a protein from amino acids requires energy.

188. Answer (1)

Hint: Does not represent 3-D view of proteins

Sol.: In proteins, only right handed helices are observed. Some regions of the proteins thread are folded into forms in what is called the secondary structure.

Haemoglobin represents the quaternary structure of protein.

189. Answer (4)

Hint: Yellow collagen fibres

Sol.: In all connective tissues, except blood, the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissue. These cells and fibres act as the matrix (ground substance).

190. Answer (3)

Hint: Exclude examples of dense regular connective tissues

Sol.: The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys and its main function is secretion and absorption.

- Tendons attach skeletal muscles to bones while ligaments attach one bone to another.
- Neurons possess axon, dendrites as well as cyton but not neuroglia.

191. Answer (4)

Hint: Involuntary muscles

Sol.: Cardiac muscle fibres and smooth muscle fibres are involuntary muscle fibres. Skeletal muscle fibres are voluntary muscle fibres.

192. Answer (1)

Hint: More than the C-atoms in palmitic acid

Sol.: Palmitic acid has 16 carbons including the carboxyl carbon. Arachidonic acid has 20 carbon atoms including the carboxyl carbon.

193. Answer (2)

Hint: Essential amino acid

Sol.: Humans are incapable of synthesizing half of the 20 standard amino acids and these are known as essential amino acids. Thus, they must be obtained from food. *e.g.*, Lysine, methionine, phenylalanine, tryptophan, valine, isoleucine, leucine and threonine.

Tryptophan, phenylalanine and tyrosine are aromatic amino acids.

194. Answer (2)

Hint: In competitive inhibition, K_m increases

Sol.: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibition. Due to its close structural similarity with the substrate, the inhibitor competes with the substrate for the substrate binding site of the enzyme. Consequently, the substrate cannot bind and as a result, the enzyme action declines.

195. Answer (3)

Hint: Follows Chargaff's rule

Sol.: According to Chargaff's rule,

Amount of adenine = amount of thymine

Amount of guanine = amount of cytosine

$\because$ % of guanine = 18%

$\therefore$ % cytosine = 18%

% of adenine = % of thymine = 32%

196. Answer (4)

Hint: Anthocyanins and carotenoids belong to the same category under secondary metabolites

Sol.: Chitin – Exoskeleton of arthropods

Collagen – Intercellular ground substance

Anthocyanins – Pigments

Strychnine – Produced by plants for their defence

197. Answer (2)

Hint: Also lines the oviducts

Sol.: Ciliated epithelium are mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes. Their function is to move particles or mucus in a specific direction over the epithelium.

- Brush bordered cuboidal epithelium is present in PCT.
- Germinal epithelium assists in the production of gametes.

198. Answer (3)

Hint: Produce mucus

Sol.: Unicellular glands consist of isolated glandular cells (goblet cells of the alimentary canal), and multicellular glands consist of cluster of cells (salivary gland, sweat gland, mammary gland, *etc*.)

199. Answer (1)

Hint: Three kinds of co-factors may be identified

Sol.: Enzymes are composed of one or several polypeptide chains. However, there are a number of cases in which non-protein constituents called co-factors are bound to the enzyme to make the enzyme catalytically active. In these instances, the protein portion of the enzymes is called the apoenzyme.

Holoenzyme is a complete functional enzyme, which is catalytically active.

200. Answer (2)

Hint: Communication junction

Sol.: Different types of cell junctions are found in epithelium and other tissues. Tight junctions help to stop substances from leaking across a tissue. Adhering junction performs cementing to keep neighbouring cells together. Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

Desmosomes serve anchoring function.

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